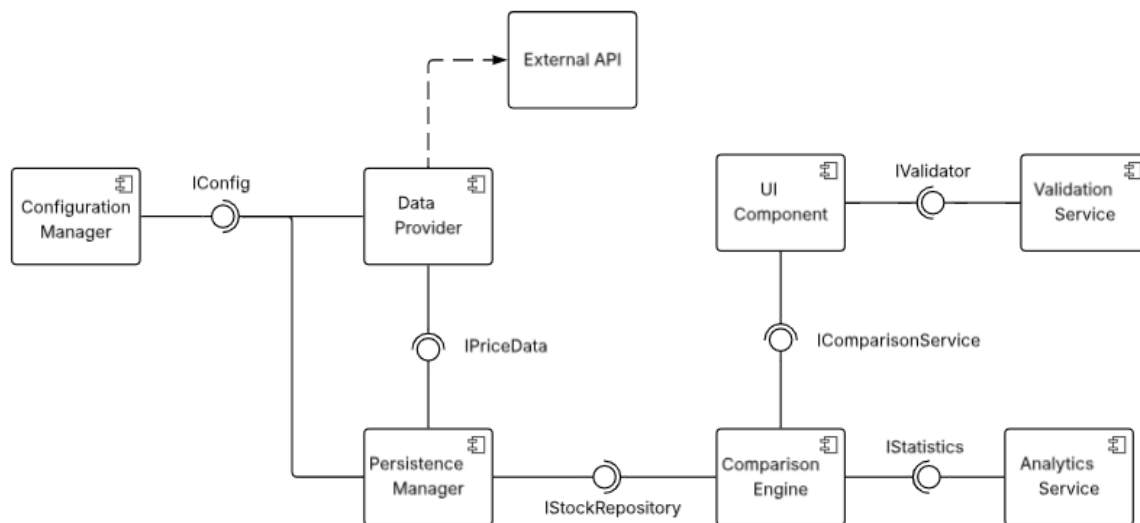




Component description

This is a Component Specification diagram, that includes four components which are 'Persistence Manager', 'Data Provider', 'UI Component', 'Configuration Manager', 'Validation Service', 'Analytics Service', and 'Comparison Engine'. The 'Data Provider' is responsible for obtaining the daily price information up to date stock market rates like Yahoo Finance. The 'Persistence Manager' handles storing and retrieving data locally to ensure its operational without a network connection. The 'Comparison Engine' is the logic that process share prices for one or two companies to prepare them for graphing. The 'UI Component' displays the share performance graphs to the user. The 'Configuration Manager' manages system-wide settings, such as API keys for stock market tracker like Yahoo Finance or the file path of our local JSON storage. The 'Validation Service' component mentions the specific constraints like a maximum range of two years for share symbols. Lastly the 'Analytics Service' seperates the mathematical calculations from the graphing logic.

Interfaces description

There are three interfaces. The 'IPriceData' sends fetched share data to storage. The 'IStockRepository' allows the engine to load data even if offline. The 'IComparisonService' provides the final graph data for display. 'Iconfig' interface supplies settings to the 'Data Provider' component to find where the external API is and 'Persitance Manager' to find out where to save local files. IValidator interface is used to check if the dates entered by the user are valid before the system attempts to fetch data. 'IStatistics' interface calculates the percentage changes, measures of rate and magnitude of price movements(volatility) and moving averages.

External dependency description

The 'External API' provides the data for the 'Data Provider' component to access.